CRAFTNCODE 2.0 • OPEN INNOVATION

Helix Reader

Turning digital reading from passive scrolling into active learning.

FOCUS MODE

VOCAB LOCK

TTS

XP + BADGES

TEAM NAME

Binary Ghosts

02 / THE WHY

Problem Statement

Digital reading gives us access — but not necessarily understanding.

Long-formdigital contentencouragesskimming, vocabularygaps, fragmentedworkflowsand distraction.Readers often jump between a book, dictionary, text-to-speech tool and revision system instead of staying inside one focused environment.

SKIMMING

Fastscanningand eye-regression make sustained attention harder, especially in long-form digital text.

VOCABULARY

A wordlookup isuseful in the moment, but it is easily forgotten because it is disconnected from revision.

FRAGMENTATION

Reading, lookup, audio,notes and progress tracking are commonly spread across separate tools.

03 / THE ANSWER

Our Solution

Helix Reader turns the reader interface into a learning system.

Rather than requiring the reader to manage multiple tools, the experience integrates support directly into the reading flow.

01 FOCUS

Guided reading views and pacing reduce visual wandering and encourage deliberate reading.

02 UNDERSTAND

Instant dictionary support, vocabulary capture and narration help readers resolve friction immediately.

03 RECALL

Saved vocabulary, quizzes, XP, badges and progress history turn reading into an active loop.

04 IMPROVE

Accessibilitycontrols and reading insights help users build a sustainable reading habit.

04 / WHAT WE BUILT

Key Features

Focus Mode

Guidedline/reading pacing to reduce visual distraction.

Vocabulary Lock

Savedifficultwordsdirectly while reading for later revision.

Text-to-Speech

Listentocontent whilemaintaining a reading-focused workflow.

Accessibility

Typography, spacing and reading controls designed for different reading needs.

Gamification

XP,streaksandbadges provide visible progress and motivation.

Reading Insights

Trackhabitsandprogress instead of treating every session as isolated.

05 / EXPERIENCE FLOW

User Journey

From opening a book to remembering what was read.

01

DISCOVER

Find or import reading material.

02

FOCUS

Enter a distraction-light reading view.

03

INTERACT

Look up words, listen, highlight and save.

04

RECALL

Use vocabulary and comprehension activities.

05

TRACK

Earn progress, streaks and reading insights.

THE HELIX LOOP Focus → Understand → Recall → Progress → Better next session

06 / HOW IT WORKS

Technical Approach

FRONTEND

HTML• CSS•JavaScript Responsive reading interface Interactive controls Focus and accessibility views Reading progress UI

BACKEND / UTILITIES

Python NLP utilities Text processing Reading-support logic Streamlit dashboard / utilities

DATA + DEPLOYMENT

SQLite User/progress stateVocabulary records Reading history Static web deployment + project repository

Methodology: modular features → reusable reading components → progressive enhancement

07 / TECHNICAL FLOW

System Architecture

CONTENT	Books • articles • imported text • external sources
READING ENGINE	Reader UI • focus pacing • highlights • TTS • dictionary
LEARNING LAYER	Vocabulary • quizzes • XP • badges • progress
DATA LAYER	Local records • reading state • vocabulary history • insights

The architecture keeps learning features around the reading experience — not in a separate app.

08 / ENGINEERING REALITY

Challenges & Risks

ACCESSIBILITY WITHOUT CLUTTER

Controls need to remaindiscoverable without making the reader interface overwhelming.

STATE & PROGRESS

Readingposition,vocabulary,highlights and achievements must stay consistent across sessions.

EXTERNAL CONTENT

Different sourcesand formatscan behave differently; fallbacks are needed to keep the workflow simple.

PERFORMANCE

Search,audio,NLPutilities and interactive UI should remain responsive as usage grows.

AI / NLP RELIABILITY

Supportive outputs need safefallbacks so uncertain results do not interrupt reading.

ENGAGEMENT BALANCE

XPand badges shouldreinforcereading habits instead of becoming another distraction.

09 / VALUE

Impact, Benefits & Future Scope

FOR READERS

Acalmer readingworkflow with immediate support, vocabulary capture, narration, accessibility controls and visible progress.

FOR EDUCATORS

Afoundationfor measurable reading habits, comprehension checkpoints and future learning-dashboard integration.

FOR INSTITUTIONS

Amodularreadingplatformthat can grow into LMS integrations and broader digital-learning workflows.

Phase1

Universal Web Clipper

Bring guided pacing to web articles.

Phase 2

personalised audio

Multi-speaker, higher-fidelity narration.

Phase 3

INSTITUTIONAL LMS

Connectprogresstoinstitutional dashboards.

RESEARCH • PROTOTYPE • OPEN INNOVATION

Thank you.

Read slower.Understanddeeper.Remember longer.

https://hardik-arora.github.io/helix-reader/

Research / integration references

Project Gutnberg • Open Library • Google Books • Internet Archive • Helix Reader repository